

PROJECT ABSTRACT

ReliefLink – Flood Relief Item Availability Portal

1. Problem Statement (Existing System)

During floods and other natural disasters, affected people often struggle to find nearby relief centers and essential supplies. Information regarding the availability of groceries, drinking water, medicines, clothes, sanitary products, baby essentials, and other relief materials is usually shared through social media, phone calls, or messaging apps, making it difficult to verify and access. Relief organizations also face challenges in managing donations and distributing resources efficiently. The lack of a centralized system results in delayed assistance, duplicate requests, and unequal distribution of relief materials.

2. Objective

The objective of ReliefLink is to develop a centralized platform that connects NGOs, relief centers, volunteers, donors, and affected people during flood emergencies. The system enables relief organizations to update available relief items in real time and allows users to check item availability before visiting a relief center. Users can reserve required items by providing their basic details, while volunteers assist in managing distributions. The platform aims to improve coordination, reduce resource wastage, and ensure faster and more transparent disaster relief management.

3. Proposed Solution

ReliefLink is a mobile application designed to simplify flood relief management. NGOs and relief centers can register and maintain the availability of essential items such as groceries, drinking water, clothes, medicines, sanitary products, baby care items, blankets, and hygiene kits. Affected users can browse nearby relief centers, view available items, and reserve required supplies by entering their name and phone number. Volunteers can verify bookings, distribute relief materials, and update delivery status. Admin manages users, NGOs, volunteers, relief centers, inventory, and bookings through a centralized dashboard. Notifications keep users informed about booking confirmations and stock updates, ensuring better coordination during emergency situations.

4. Technology Stack

Component	Technology
Frontend (Mobile)	Flutter
Backend	Node.js + Express.js
Database	PostgreSQL
Image Storage	Cloudinary
Authentication	JWT (JSON Web Token)
Notifications	Firebase Cloud Messaging (FCM)

5. Benefits of the Project

- Provides real-time information about available relief materials.
- Reduces duplicate requests and improves resource distribution.
- Enables easy reservation of relief items by affected people.
- Improves coordination among NGOs, volunteers, and relief centers.
- Ensures transparent and organized disaster relief management.

MODULE DESCRIPTION

Admin

- Manage NGOs
 - Manage Volunteers
 - Manage Users
 - Manage Relief Centers
 - View Reports
 - Manage Bookings
 - Manage Categories
 - Manage Inventory
-

NGO / Relief Center

- Register/Login
 - Create Relief Center
 - Add Available Items
 - Update Item Stock
 - Accept Bookings
 - View Reports
-

Volunteer

- Register/Login
 - View Assigned Bookings
 - Verify Distribution
 - Update Delivery Status
-

User (Affected Person)

- Register/Login
- Browse Relief Centers
- Search Available Items
- Reserve Items & view booking status
- Receive Notifications

Main Modules

1. Authentication Module
2. User Management
3. NGO / Relief Center Management
4. Volunteer Management
5. Relief Item Management
6. Inventory Management
7. Item Booking System
8. Notifications
9. Reports & Analytics
10. Admin Dashboard

Total Modules: 10